

PROBLEM STATEMENT: KPI'S REQUIREMENT

We need to analyze key indicators for our pizza sales data to gain insights into our business performance. Specifically, we want to calculate the following metrics:

1. **Total Revenue:** The sum of the total price of all pizza orders.
2. **Average Order Value:** The average amount spent per order, calculated by dividing the total revenue by the total number of orders.
3. **Total Pizzas Sold:** The sum of the quantities of all pizzas sold.
4. **Total Orders:** The total number of orders placed.
5. **Average Pizzas Per Order:** The average number of pizzas sold per order, calculated by dividing the total number of pizzas sold by the total number of orders.

PROBLEM STATEMENT: CHARTS REQUIREMENT

We would like to visualize various aspects of our pizza sales data to gain insights and understand key trends. We have identified the following requirements for creating charts:

1. **Daily Trend for Total Orders:** Create a bar chart that displays the daily trend of total orders over a specific time period. This chart will help us identify any patterns or fluctuations in order volumes on a daily basis.
2. **Monthly Trend for Total Orders:** Create a line chart that illustrates the hourly trend of total orders throughout the day. This chart will allow us identify peak hours or periods of high order activity. *(Note: While labeled "Monthly Trend," the description specifies "hourly trend throughout the day").*
3. **Percentage of Sales by Pizza Category:** Create a pie chart that shows the distribution of sales across different pizza categories. This chart will

provide insights into the popularity of various pizza categories and their contribution to overall sales.

4. **Percentage of Sales by Pizza Size:** Generate a pie chart that represents the percentage of sales attributed to different pizza sizes. This chart will help us understand customer preferences for pizza sizes and their impact on sales.
5. **Total Pizzas Sold by Pizza Category:** Create a funnel chart that presents the total number of pizzas sold for each pizza category. This chart will allow us to compare the sales performance of different pizza categories.
6. **Top 5 Best Sellers by Revenue, Total Quantity and Total Orders:** Create a bar chart highlighting the top 5 best-selling pizzas based on the Revenue, Total Quantity, Total Orders. This chart will help us identify the most popular pizza options.
7. **Bottom 5 Best Sellers by Revenue, Total Quantity and Total Orders:** Create a bar chart showcasing the bottom 5 worst-selling pizzas based on the Revenue, Total Quantity, Total Orders. This chart will enable us to identify underperforming or less popular pizza options.

Approach

◆ Step 1: Data Collection

- Dataset used: pizza_sales.csv
 - Contains:
 - Order ID
 - Pizza Name
 - Category (Classic, Veggie, Chicken, Supreme)
 - Size (S, M, L, XL)
 - Quantity
 - Price
 - Date & Time
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◆ Step 2: Data Cleaning (Power BI)

- Removed null values
- Corrected data types (date, number)
- Created new columns:
 - Day Name
 - Month Name
 - Total Revenue = Quantity × Price

◆ Step 3: Data Modeling

- Used single dataset (no complex relationships)
- **Created calculated measures:**
 - Total Revenue

- Total Orders
 - Total Quantity Sold
 - Avg Order Value
 - Avg Pizza per Order
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◆ Step 4: Dashboard Design

Overview Dashboard

- **KPI Cards:**
 - Total Revenue → 817.86K
 - Total Orders → 21,350
 - Total Pizza Sold → 49,574
 - Avg Order Value → 38.31
 - Avg Pizza/Order → 2.32
- **Visuals:**
 - Daily Orders Trend
 - Monthly Orders Trend
 - Category-wise Sales %
 - Size-wise Sales %
 - Sales by Category

Best & Worst Sellers

- **Top 5 pizzas by:**
 - Revenue
 - Quantity

- Orders
- **Bottom 5 pizzas by:**
 - Revenue
 - Quantity
 - Orders

◆ **Interactivity**

- **Filters used:**
 - Pizza Category
 - Date Range Slider
- **Navigation buttons:**
 - Home
 - Best/Worst Seller

Key Insights

Sales Insights

- **Classic Category** generates maximum revenue & orders
 - **Large Size pizzas** contribute the most sales (~45%)
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Time-Based Insights

- Orders are highest on:
 - **Weekends (Friday & Saturday)**
 - Peak months:
 - **July & January**
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Product Insights

- **Best Seller:**
 - Thai Chicken Pizza (Revenue)
 - Classic Deluxe Pizza (Orders & Quantity)
 - **Worst Seller:**
 - Brie Carre Pizza (lowest across all metrics)
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Customer Behavior

- **Average:**
 - **2.32 pizzas per order**
- **Indicates:**
 - Customers prefer ordering multiple items

Data Source

This is:

 **Not Live Data**

 **Static Dataset (CSV File)**

Explanation:

- Data is imported from a file
- No real-time connection
- Dashboard updates only when data is refreshed manually

If needed, it can be converted to:

- SQL Server (Live)
- Azure Database (Cloud Live Data)

Recommendations

Business Improvements

1. Focus on Best Sellers

- Promote:
 - Thai Chicken Pizza
 - Classic Deluxe Pizza
- Use combo offers

2. Improve Worst Sellers

- Re-evaluate:
 - Pricing
 - Ingredients
- Offer discounts or remove from menu

3. Weekend Strategy

- Increase:
 - Staff
 - Inventory
- Run weekend offers

4. Upselling Strategy

- Since avg = 2.32 pizzas/order:
 - Offer “Buy 2 Get 1”
 - Add beverages/combo deals

5. Seasonal Marketing

- Focus ads on:
 - July
 - January

6. Size Optimization

- Promote:
 - Large size pizzas (highest demand)

